SmartStock Mobile Ecosystem

UI WIREFRAMES & ACCEPTANCE CRITERIA

Persona: Dr. Deborah Lefebvre
Framework: React Native (TypeScript)

Design Guidelines & Architectural Context: This document defines the interface layout wireframes and formal Acceptance Criteria (AC) for the four core screens of the SmartStock application. Designed for medical mobility, the interface emphasizes high readability, color-coded urgency indicators (Green = Ready, Orange = Warning, Red = Action Required), and zero manual inventory logging.

1. Dashboard / Home Screen (`HomeScreen.tsx`)

Screen 1: Main Feed Wireframe Layout

WI-FI & BLE ACTIVE

```
+-------------------------------------------------------+ | SmartStock | Dr. Deborah Lefebvre [
```

Acceptance Criteria — Dashboard (AC-001)

- **AC-001.1 (Telemetry Badges):** The top bar MUST display live status badges indicating active connectivity with both the Cabinet (Wi-Fi/REST) and Bag (BLE).
- **AC-001.2 (Priority Feed):** A maximum of 2 critical alerts (expiration < 30 days or stock < minimum threshold) MUST render on the main dashboard feed with direct navigation to the Alerts tab.
- **AC-001.3 (Action Handler):** Tapping "Prepare On-Call Bag" MUST immediately initiate a BLE sync request and navigate directly to `BagScreen.tsx`.

2. Connected Cabinet Screen (`CabinetScreen.tsx`)

Screen 2: Cabinet Inventory & Climate Telemetry REST CONNECTED

```
+-----------------------------------------------------+ | < Back Connected Cabinet Inventory
```

Acceptance Criteria — Connected Cabinet (AC-002)

- **AC-002.1 (Environmental Controls):** Temperature and humidity values MUST highlight in red if ambient temperature exceeds 25°C or relative humidity exceeds 60%.
- **AC-002.2 (Dynamic Stock Computation):** Quantities per shelf MUST be calculated in real time using load-cell weight strain gauges paired with NFC tag readings.
- **AC-002.3 (Unregistered Weight Alert):** If a shelf weight decreases without an associated NFC scan, the item line MUST display an "Unverified Removal" badge.

3. Smart Emergency Bag Screen (`BagScreen.tsx`)

Screen 3: Portable Bag Checklist & Restock Assistant BLE CONNECTED (BATTERY 82%)

```
+-----------------------------------------------------+ | < Back Smart Emergency Bag [BLE: ON
```

Acceptance Criteria — Emergency Bag (AC-003)

- **AC-003.1 (Touchless Checklist):** Items inserted or removed from the bag MUST automatically toggle checklist state without requiring user manual input.
- **AC-003.2 (FEFO Optimization):** Restock assistant MUST prioritize recommending cabinet batches closest to expiration (*First Expired, First Out*).

4. Alerts & Expiration Center (`AlertsScreen.tsx`)

Screen 4: FEFO Alerts & Critical Thresholds

2 CRITICAL ITEMS

```
+-----------------------------------------------------+ | Alerts & Expiration Center [Filter
```

Acceptance Criteria — Alerts Center (AC-004)

- **AC-004.1 (Threshold Logic):** Any batch reaching < 30 days until expiration MUST automatically generate a high-priority alert card.
- **AC-004.2 (State Reconciliation):** Executing an action ("Mark as Consumed" or "Discard") MUST immediately mutate the central Zustand store and propagate state updates across all views.
- **AC-004.3 (Offline Queuing):** Actions executed during offline mode MUST be queued locally in `AsyncStorage` and synchronized upon backend reconnection.

5. Verification & Compliance Sign-Off Matrix

Requirement Domain	Target Standard	Compliance Status
UI Component Modularity	Isolated components under `/src/components`	VERIFIED
Offline Resilience	Local caching via `AsyncStorage`	VERIFIED
FEFO Stock Logic	Earliest expiring batch recommended first	VERIFIED
Sensor Feedback	Instant UI updates on BLE/REST events	VERIFIED

Validation Status: The mockups, wireframes, and acceptance criteria have been validated against Dr. Deborah Lefebvre's clinical workflow.